

Skills Summary

Ordering Rational and Irrational Numbers

Use this reference while completing your worksheet or when studying.

Skill 1 — Putting every quantity into decimal form

Rule: numbers can only be compared in the same form, and the decimal is the form everything converts to.

Fraction: divide top by bottom — the result terminates or repeats, exactly. **Root:** squeeze between decimals whose squares trap the radicand, and round. **Pi:** use three decimal places when a comparison is close.

Example: Write $\frac{7}{8}$ and $\sqrt{13}$ as decimals.

Step 1. One is a fraction and one is a root, so they need different tools — division for the fraction, a squeeze for the root.

Step 2. Dividing seven by eight terminates after three places; for the root, the squares of three point six and three point seven straddle the radicand, so squeeze once more with hundredths.

$$\frac{7}{8} = 0.875 \quad \text{and} \quad \sqrt{13} \approx 3.61$$

Try it: Write $\frac{5}{16}$ and $\sqrt{23}$ as decimals.

check: 0.3125 and 4.80

Skill 2 — Comparing two quantities

Method: convert both, then compare place by place from the left. Keep one more decimal place than the point where they first differ.

Negatives reverse: further left on the number line is *smaller*, so among negatives the one with the bigger digits is the smaller number.

Example: Write $<$, $>$, or $=$ between $\frac{5}{7}$ and 0.72.

Step 1. A fraction against a decimal cannot be read off as printed, so convert the fraction and line the two decimals up.

Step 2. Dividing five by seven repeats, and it first differs from the given decimal in the hundredths place, where it is the smaller of the two.

$$\Rightarrow \frac{5}{7} < 0.72$$

Try it: Write $<$, $>$, or $=$ between $\sqrt{8}$ and 2.9.

check: $>$

(!) **Watch out:** do not judge size from how a number is printed. A short-looking root can be larger than a long-looking decimal, and a bigger denominator does not mean a bigger number. Convert first, every time.

Skill 3 — Ordering a mixed list

Method: convert every entry to the same number of decimal places, write the decimals underneath one another, order those, then rewrite the answer using the *original* forms — that is what the question asked for.

Example: Order $\sqrt{7}$, 2.6 and $\frac{8}{3}$, least to greatest.

Step 1. Three different forms, so nothing can be ordered until all three are decimals at the same precision.

Step 2. The root squeezes to a little under two point sixty-five, the fraction repeats a little over two point sixty-six, and the plain decimal is the smallest of the three.

$$\Rightarrow 2.6 < \sqrt{7} < \frac{8}{3}$$

Try it: Order $\sqrt{3}$, 1.8 and $\frac{7}{4}$, least to greatest.

check: $1.8 < \sqrt{3} < \frac{7}{4}$